Geometry & Measurement

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Triangle angle sum**

Find the third angle of the triangle shown, given the two marked angles.

$180 - 40 - 75 = 65^\circ$.

2 **[Foundational]** • **Target: The Pythagorean theorem**

Find the hypotenuse of the right triangle shown.

$c = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10$ cm.

3 **[Foundational]** • **Target: Circles: circumference and area**

Find the area of the shaded 90° sector shown, in terms of π.

Sector area $= \frac{90}{360} \times \pi \times 10^2 = \frac{1}{4} \times 100\pi = 25\pi$ cm^2.

4 **[Foundational]** • **Target: Composite figures**

Find the area of the L-shaped room shown.

Treat it as a 10 m $\times$ 8 m rectangle with a 4 m $\times$ 3 m notch removed: $10 \times 8 - 4 \times 3 = 80 - 12 = 68$ m^2.

5 **[Foundational]** • **Target: Angle facts**

Two angles on a straight line measure $3x$ and $2x + 30$. Find x.

Angles on a straight line sum to 180°: $3x + 2x + 30 = 180 \Rightarrow 5x = 150 \Rightarrow x = 30$.

6 **[Foundational]** • **Target: Angle facts**

Find the angle vertically opposite a 65° angle.

Vertically opposite angles are equal, so the angle is 65°.

7 **[Foundational]** • **Target: Angle facts**

Three angles around a point measure 110°, 95°, and x. Find x.

Angles around a point sum to 360°: $x = 360 - 110 - 95 = 155^\circ$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Angle facts

Two angles are complementary (sum to 90°). One measures 37°. Find the other.

$90 - 37 = 53^\circ$.

9 [Intermediate] • Target: Parallel lines and transversals

Two parallel lines are cut by a transversal, creating an angle of 70°. Find its corresponding angle.

Corresponding angles are equal, so the angle is 70°.

10 [Intermediate] • Target: Parallel lines and transversals

In the same setup, find the co-interior (allied) angle to a 70° angle.

Co-interior angles sum to 180°: $180 - 70 = 110^\circ$.

11 [Intermediate] • Target: Parallel lines and transversals

Find the alternate interior angle to a 55° angle.

Alternate interior angles are equal, so the angle is 55°.

12 [Intermediate] • Target: Parallel lines and transversals

Two parallel lines cut by a transversal create an angle of $4x$ and its co-interior angle of $2x + 30$. Find x.

Co-interior angles sum to 180°: $4x + 2x + 30 = 180 \Rightarrow 6x = 150 \Rightarrow x = 25$.

13 [Intermediate] • Target: Triangle angle sum

An isosceles triangle has two equal angles of 72° each. Find the third angle.

$180 - 72 - 72 = 36^\circ$.

14 [Intermediate] • Target: Triangle angle sum

In a right triangle, one of the non-right angles is 35°. Find the other non-right angle.

$180 - 90 - 35 = 55^\circ$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Triangle angle sum • Cross-Lesson: 1.01 Arithmetic & Number Sense

A triangle has three equal angles. Find each angle, and name this type of triangle.

$180 \div 3 = 60^\circ$ each — this is an equilateral (equiangular) triangle.

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16 [Challenging] • Target: Angle sums of polygons • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Find the sum of the interior angles of a hexagon (6 sides).

Sum $= (n-2) \times 180^\circ = (6-2) \times 180^\circ = 720^\circ$.

17 [Challenging] • Target: Angle sums of polygons • Cross-Lesson: 3.01 Algebra Essentials

Find the sum of the interior angles of a pentagon (5 sides).

Sum $= (5-2) \times 180^\circ = 540^\circ$.

18 [Challenging] • Target: Angle sums of polygons • Cross-Lesson: 5.01 Quantitative Comparisons

A regular polygon has an interior angle sum of $1{,}080^\circ$. How many sides does it have?

$(n-2) \times 180 = 1{,}080 \Rightarrow n-2 = 6 \Rightarrow n = 8$ sides (a regular octagon).

19 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 6.01 Sequences & Patterns

A right triangle has hypotenuse 13 and one leg 5. Find the other leg.

$\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$.

20 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 7.01 Data Analysis & Probability

A 10 m ladder leans against a wall with its foot 6 m from the wall's base. How high up the wall does it reach?

$\sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8$ m.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Triangle angle sum

Find the third angle of the triangle shown.

a. 116°

b. 64°

c. 54°

d. 44° ✓

$180 - 64 - 72 = 44^\circ$.

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22 [Foundational] • Target: MCQ — Triangle angle sum

The isosceles triangle shown has two equal base angles of 70° each. Find the apex angle.

a. 55°

b. 70°

c. 40° ✓

d. 110°

$180 - 70 - 70 = 40^\circ$.

23 [Foundational] • Target: MCQ — The Pythagorean theorem

Find the missing leg of the right triangle shown.

a. 26

b. 18

c. 24 ✓

d. 20

$\sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$.

24 [Foundational] • Target: MCQ — The Pythagorean theorem

A triangle has sides 9 cm, 12 cm, and 15 cm, as shown. Is it a right triangle?

a. Only if it is also isosceles

b. **Yes — a right triangle** ✓

c. Cannot be determined

d. No — not a right triangle

$9^2 + 12^2 = 81 + 144 = 225 = 15^2$, so yes, it is a right triangle.

25 [Foundational] • Target: MCQ — Perimeter and area of rectangles and squares

Find the perimeter and area of the rectangle shown.

a. $P = 48$ **cm,** $A = 135$ **cm**^{2} ✓

b. $P = 48$ cm, $A = 24$ cm^2

c. $P = 135$ cm, $A = 48$ cm^2

d. $P = 24$ cm, $A = 135$ cm^2

$P = 2(15 + 9) = 48$ cm; $A = 15 \times 9 = 135$ cm^2.

26 [Foundational] • Target: MCQ — Perimeter and area of rectangles and squares

Find the perimeter and area of the square shown.

a. $P = 44$ cm, $A = 110$ cm^2

b. $P = 44$ **cm,** $A = 121$ **cm**^{2} ✓

c. $P = 22$ cm, $A = 121$ cm^2

d. $P = 121$ cm, $A = 44$ cm^2

$P = 4 \times 11 = 44$ cm; $A = 11^2 = 121$ cm^2.

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27 [Foundational] • Target: MCQ — Area of triangles and parallelograms

The triangle shown has base 16 cm and height 10 cm. Find its area.

a. 160

b. 26

c. 80 ✓

d. 40

$A = \frac{1}{2}(16)(10) = 80$ cm^2.

28 [Foundational] • Target: MCQ — Area of triangles and parallelograms

The parallelogram shown has base 18 cm and height 7 cm. Find its area.

a. 252

b. 126 ✓

c. 63

d. 25

$A = 18 \times 7 = 126$ cm^2.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Angle facts

Two angles are supplementary. One is three times the other. Find the smaller angle.

a. 60°

b. 30°

c. 135°

d. 45° ✓

$x + 3x = 180 \Rightarrow x = 45^\circ$.

30 [Intermediate] • Target: MCQ — Parallel lines and transversals

Two parallel lines are cut by a transversal. One angle is $(3x + 15)^\circ$ and its corresponding angle is $(5x - 25)^\circ$. Find x.

a. 10

b. 8

c. 30

d. 20 ✓

Corresponding angles are equal: $3x + 15 = 5x - 25 \Rightarrow x = 20$.

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31 [Intermediate] • Target: MCQ — Triangle angle sum

A triangle has three angles in the ratio $2:3:4$. Find the largest angle.

a. 90°

b. 100°

c. 80° ✓

d. 60°

9 parts $= 180^\circ$, so 1 part $= 20^\circ$; largest $= 4 \times 20 = 80^\circ$.

32 [Intermediate] • Target: MCQ — Perimeter and area of rectangles and squares

A square has area 225 cm^2. Find its side length and perimeter.

a. $s = 15$ cm, $P = 30$ cm

b. $s = 225$ cm, $P = 900$ cm

c. $s = 15$ **cm,** $P = 60$ **cm** ✓

d. $s = 45$ cm, $P = 60$ cm

$s = \sqrt{225} = 15$ cm; $P = 4 \times 15 = 60$ cm.

33 [Intermediate] • Target: MCQ — Composite figures

A garden is L-shaped: an 18 m by 8 m rectangle with a 6 m by 3 m rectangle removed from one corner. Find its area.

a. 108

b. 126 ✓

c. 162

d. 138

$18 \times 8 = 144$; $144 - 6 \times 3 = 144 - 18 = 126$ m^2.

34 [Intermediate] • Target: MCQ — Similar figures and scale factor

Two similar triangles have a scale factor of $3:5$. If the smaller triangle's perimeter is 24 cm, find the larger triangle's perimeter.

a. 14.4

b. 72

c. 64

d. 40 ✓

$24 \times \frac{5}{3} = 40$ cm.

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35 [Intermediate] • Target: MCQ — Word problems

Tiling costs 38 riyals per square meter. Find the total cost to tile a rectangular room measuring 9 m by 6 m.

a. 1,026 SAR

b. 4,104 SAR

c. 342 SAR

d. 2,052 **SAR** ✓

Area $= 9 \times 6 = 54$ m^2; cost $= 54 \times 38 = 2{,}052$ SAR.

36 [Intermediate] • Target: MCQ — Perimeter and area of rectangles and squares

A right triangle has legs 33 cm and 180 cm. Find the hypotenuse.

a. 183 ✓

b. 181

c. 213

d. 185

$c = \sqrt{33^2 + 180^2} = \sqrt{1089 + 32400} = \sqrt{33489} = 183$ cm.

37 [Intermediate] • Target: MCQ — Area of triangles and parallelograms

A right triangle has legs 20 cm and 48 cm. Find the hypotenuse.

a. 54

b. 52 ✓

c. 50

d. 68

$c = \sqrt{20^2 + 48^2} = \sqrt{400 + 2304} = \sqrt{2704} = 52$ cm.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Circles: circumference and area • Cross-Lesson: 5.01 Quantitative Comparisons

A right triangle has legs 27 cm and 120 cm. Find the hypotenuse.

a. 121

b. 125

c. 123 ✓

d. 147

$c = \sqrt{27^2 + 120^2} = \sqrt{729 + 14400} = \sqrt{15129} = 123$ cm.

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39 [Challenging] • Target: MCQ — Composite figures • Cross-Lesson: 6.01 Sequences & Patterns

A rectangle has length 9 cm and width 8 cm. Find its perimeter and area.

a. $34, 78$

b. $38, 72$

c. $34, 72$ ✓

d. $72, 34$

Perimeter $= 2(9 + 8) = 34$ cm. Area $= 9 \times 8 = 72$ cm^2.

40 [Challenging] • Target: MCQ — Volume and surface area • Cross-Lesson: 7.01 Data Analysis & Probability

A rectangle has length 6 cm and width 4 cm. Find its perimeter and area.

a. $24, 24$

b. $20, 24$ ✓

c. $20, 30$

d. $24, 20$

Perimeter $= 2(6 + 4) = 20$ cm. Area $= 6 \times 4 = 24$ cm^2.

41 [Challenging] • Target: MCQ — Coordinate geometry: distance and midpoint • Cross-Lesson: 8.01 Word Problems & Reasoning

A rectangle has length 12 cm and width 6 cm. Find its perimeter and area.

a. $36, 72$ ✓

b. $36, 78$

c. $72, 36$

d. $40, 72$

Perimeter $= 2(12 + 6) = 36$ cm. Area $= 12 \times 6 = 72$ cm^2.

42 [Challenging] • Target: MCQ — Similar figures and scale factor • Cross-Lesson: 1.01 Arithmetic & Number Sense

A rectangle has length 21 cm and width 14 cm. Find its perimeter and area.

a. $294, 70$

b. $70, 294$ ✓

c. $70, 300$

d. $74, 294$

Perimeter $= 2(21 + 14) = 70$ cm. Area $= 21 \times 14 = 294$ cm^2.

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43 [Challenging] • Target: MCQ — Word problems • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

An isosceles triangle has two equal angles of 64° each. Find the third angle.

a. 52 ✓

b. 46

c. 64

d. 58

$180 - 64 - 64 = 52^\circ$.

44 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Algebra Essentials

Find the sum of the interior angles of an octagon (8 sides).

a. 1080 ✓

b. 1440

c. 1260

d. 900

Sum $= (n-2) \times 180^\circ = (8-2) \times 180^\circ = 1080^\circ$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Quantitative Comparisons

A circle has radius 35 cm. Find its area, using $\pi = \frac{22}{7}$.

a. 3864

b. 220

c. 3850 ✓

d. 3836

$A = \pi r^2 = \frac{22}{7} \times 1225 = 3850 \text{ cm}^2$.

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